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Pet Loss: Understanding Disenfranchised Grief, Memorial Use, and Posttraumatic Growth

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ABSTRACT The growing body of literature exploring pet loss suggests that many bereaved pet owners experience disenfranchised grief. Disenfranchised grief occurs when a loss is unacknowledged and the bereaved are unable to express their grief. When grief is considered illegitimate, the bereaved may experience a variety of adverse psychological outcomes, including increased distress and reduced quality of life. Additionally, when grief is inhibited, the individual may be less likely to experience positive changes such as posttraumatic growth. Few studies have considered the relationship between disenfranchisement and posttraumatic growth following the loss of a companion animal. This study investigated the relationship between disenfranchised grief, memorialization, and posttraumatic growth in bereaved pet owners. Grief severity was assessed as a moderator. Using an online survey, respondents ($n = 133$) completed standardized measures, including the Pet Bereavement Questionnaire, the Loss of Social Support subscale from the Grief Experience Questionnaire, and the Posttraumatic Growth Inventory-Short Form (PTGI-SF). As hypothesized, there was a significant interaction between disenfranchised grief, grief severity, and posttraumatic growth, suggesting that disenfranchised grief inhibits posttraumatic growth when grief severity is high. Contrary to the hypothesis, there was no relationship between memorial quantity or type and posttraumatic growth. Overall, this study shows that disenfranchised grief can inhibit posttraumatic growth following the loss of a pet, highlighting the complicated relationship between posttraumatic growth, grief intensity, and disenfranchised grief.

Keywords: disenfranchised grief, grief severity, human–animal interaction, memorials, pet loss, posttraumatic growth



Approximately 62% of households in Australia have a companion animal, totaling over 24 million pets across the country (Animal Medicines Australia [AMA], 2016). Human–animal bonds

are often very close. Using a representative sample of 2,022 Australian adults, the AMA found that 65% of dog owners and 66% of cat owners consider their pet to be a member of the family (AMA, 2016). Kemp, Jacobs, and Stewart's (2016) systematic review of qualitative studies on pet loss also reveals that phrases such as "baby," "loved one," and "best friend" are often used by people to describe their relationship with their pet.

According to Bowlby's (1982) attachment theory, attachment to significant (human) others provides a sense of security and safety. Sable (1995) reviewed past literature on pet attachment and concluded that human–animal bonds provide the same emotional attachment and sense of security. The human–animal bond can also provide physical and mental health benefits. For example, Allen, Blascovich, and Mendes (2002) found that the presence of a pet can buffer cardiovascular reactivity to stress, and Aydin et al. (2011) reported that dogs can reduce the adverse psychological consequences (such as distress and low self-esteem) of social exclusion. Sharkin and Knox (2003) explain these findings by saying attachment to pets provides non-judgmental social support, something which is hard to achieve in human–human relationships. The corollary is that loss of a companion animal can be extremely distressing for individuals with a close bond.

Disenfranchised Grief

Every society has a set of social norms that dictate appropriate grieving behavior. These rules determine who has a legitimate right to grieve, when and how they should grieve, and for whom they should grieve (Doka, 1999). Disenfranchised grief occurs when the loss is unacknowledged or considered insignificant. Doka (1999) identified that disenfranchised grief might occur when: (a) the relationship is not recognized, (b) the loss is not recognized, and (c) the griever is not recognized. Pet loss is typically an example of the second category, which occurs when friends, family, and the broader community do not recognize the loss of a pet as a "real loss" (Corr, 1999; Doka, 2002; Packman et al., 2014). Bereaved pet owners are often expected to resolve their grief quickly and may be encouraged to replace the deceased pet (Cordaro, 2012).

Disenfranchised grief occurs within a broader system of empathic failure, where social, psychological, and relational processes interact to inhibit social support (Doka, 2002). Neimeyer and Jordan (2002) developed a model that positions disenfranchised grief within four levels of empathic failure: (a) self with self, (b) self with family, (c) self with community, and (d) self with transcendent reality. At the first level, self-disenfranchisement occurs when the emotional reaction to the loss is minimized by the bereaved. Secondly, family, friends, and the broader community may not understand or recognize the significance of the loss (Neimeyer & Jordan, 2002). Consequently, the bereaved may not be socially supported and grief may be actively discouraged (Adams, Bonnett, & Meek, 1999; Chur-Hansen, 2010). The final level of empathic failure refers to spiritual disenfranchisement, where religious beliefs held by the bereaved or by their broader religious community invalidate grief reactions.

Disenfranchised individuals are more likely to experience a range of adverse psychological outcomes. For example, Tzivian, Friger, and Kushnir's (2015) study of quality of life in 213 current and bereaved dog owners found a significant positive correlation between quality of life and social support. In a qualitative, cross-cultural study with 235 American and French-Canadian participants who had lost a companion animal, content analysis of survey data identified themes of social support in only one to two percent of responses (Packman et al., 2014). However, 6% (American) and 14% (French Canadian) of responses related to disenfranchised

grief. The severity of grief and inability to express themselves caused some participants to become depressed and consider suicide. Further, Adrian and Stitt (2017) found that bereaved pet owners in Hawaii who were unable to grieve were more likely to experience higher levels of anxiety, depression and grief severity, and had more difficulty resolving their grief.

Rituals and Memorials

Doka (2002) argues that rituals help to establish the “right to grieve” following a loss. Therefore, the absence of socially sanctioned mourning practices disenfranchises the griever by preventing recognition of the loss, reinforcing the insignificance of the loss, and impeding social support. Castle and Phillips (2003) assessed the use of rituals following the loss of a loved one and reported that the use of ritual facilitated adjustment to the loss. Importantly, respondents rated the ability to express their emotions in a safe environment as the most helpful aspect of a ritual (Castle & Phillips, 2003).

Many Western countries have limited access to culture-specific, standardized rituals following pet loss. Bereaved pet owners can arrange for cremation and a limited range of memorialization services. However, Chur-Hansen (2011) argues that these services are a procedure, rather than a ritual. For example, many Australian providers of cremation services for pets do not offer funeral services (Chur-Hansen, 2011). Conversely, Japanese pet memorial services include many of the same rituals used following the loss of a human, and grieving practices can include the use of coffins, cremation, offerings of food and incense, and regular memorial services (Kenney, 2004).

Posttraumatic Growth

Challenging life events can cause a variety of adverse psychological consequences and outcomes. However, positive change, or posttraumatic growth, can occur alongside adverse reactions to trauma (Tedeschi & Calhoun, 2004). Growth can arise as a result of struggling with a wide variety of stressful life events, including illness, accidents, and bereavement (Calhoun, Tedeschi, Cann, & Hanks, 2010). The severity of trauma can affect growth. Morris, Shakespeare-Finch, Rieck, and Newbery (2005) found that individuals who rated a traumatic event as severe experienced more posttraumatic growth than those who gave lower ratings. This relationship remained significant when controlling for age, time since trauma, and other variables.

Posttraumatic growth can occur within several broad categories or domains. Tedeschi and Calhoun (1996) identified five factors of posttraumatic growth: Relating to Others, New Possibilities, Personal Strength, Spiritual Change, and Appreciation of Life. Growth within the Relating to Others domain is the development of closer relationships with others and increases in empathy (Calhoun et al., 2010). Growth representing New Possibilities such as major life changes, taking on new roles and responsibilities, and new interpersonal relationships may occur following trauma. Personal Strength refers to increased recognition of the ability to cope with stressful life events even when feeling more vulnerable (Tedeschi & Calhoun, 1996). Spiritual Change includes existential questioning and changes in the individual's connection to the transcendent. Finally, people may change their priorities and have an increased Appreciation of Life (Tedeschi & Calhoun, 1996).

Packman, Bussolari, Katz, Carmack, and Field (2017) assessed how bereaved pet owners' responses corresponded to Tedeschi and Calhoun's (1996) five-factor model of posttraumatic growth. Overall, they found that Relating to Others was the most endorsed theme. Participants indicated that sharing the loss with their immediate family improved their relationships. However, very few participants felt socially supported following their loss, a possible

consequence of disenfranchisement (Packman et al., 2017). In a cross-cultural study, Bussolari et al. (2017) compared posttraumatic growth following the loss of a pet across French-Canadian, United States, Japanese, and Hong Kong samples. They found that Relating to Others was the most commonly endorsed theme for the US sample but was much less prominent for all other samples. Packman et al. (2017) found strong links between the Personal Strength theme and euthanasia, an issue unique to pet loss. Many participants indicated the decision to euthanize their pet helped them realize that they were stronger than they expected. Few participants reported that their faith was strengthened by their loss, although many developed a greater understanding of spiritual matters.

The Functional-Descriptive Model

This model of posttraumatic growth posits that posttraumatic growth occurs when a traumatic event challenges the individual's *assumptive world beliefs* (Tedeschi & Calhoun, 2004). Assumptive beliefs encompass everything within an individual's experience, including past events and thoughts of the future (Parkes, 1971) and are used to guide actions, understand the world, provide explanations for events, and create a sense of purpose (Tedeschi & Calhoun, 2004). When trauma challenges the assumptive world, the individual experiences a process of rumination, re-evaluation of beliefs, and cognitive reconstruction.

Cognitive reconstruction is facilitated or inhibited by sociocultural influences, including social support, role models, responses to self-disclosure, and cultural beliefs, rituals, and norms (Calhoun et al., 2010). For example, Cobb, Tedeschi, Calhoun, and Cann (2006) examined correlates of growth in 60 women who had experienced intimate partner violence, finding that women with role models reported more growth overall than those without. Another study comparing grief processing in bereaved individuals living in the United States or China found that it and long-term adjustment varies by culture (Bonanno, Papa, Lalande, Zhang, & Noll, 2005). The Chinese sample reported more initial distress and a more substantial reduction in distress over time compared with the American sample. These differences are likely to be due to Chinese cultural norms such as high interpersonal connectedness, social support, and use of ritual following bereavement. Wong, Lau, Liu, Yuen, and Wing-Lok (2015) found that Chinese respondents who had lost a companion animal in the previous three years and had received social or professional support were more likely to experience growth than those who did not receive support.

The growing body of literature exploring pet loss shows that many bereaved pet owners experience disenfranchised grief. Many of these studies examine the adverse psychological consequences of disenfranchisement, including increased grief severity and duration, resulting in higher levels of depression and anxiety. Further, several studies have documented the impact of negative sociocultural influences on posttraumatic growth. Despite the potential inhibition of growth, few studies have directly examined predictors (such as disenfranchised grief and grief severity) of posttraumatic growth following the loss of a companion animal. Therefore, this study aimed to explore posttraumatic growth in an Australian sample of bereaved pet owners. We hypothesized that:

- 1) Disenfranchised grief and grief severity predict posttraumatic growth in bereaved pet owners
- 2) The number of memorials, type of memorials, and grief severity predict posttraumatic growth in bereaved pet owners.

Methods

Participants and Procedure

Participants were recruited to an online survey via social media, primarily the authors' Facebook friends. Additionally, psychology students at Central Queensland University were invited to participate. We deliberately did not target online pet bereavement groups to avoid bias toward individuals with unusually high levels of (enduring) grief. The recruitment message stated that eligible participants were required to be at least 18 years of age, living in Australia, and have lost a pet in the last five years (but more than three months before participation). This timeframe allowed enough time for posttraumatic growth to develop (Cann et al., 2010a), but not enough time for growth to decline (Wolchik, Cox, Tein, Sandler, & Ayers, 2009). Participants were provided with information explaining the goals, benefits, and risks of the study on the first page of the online survey. For participants experiencing distress, contact details for Lifeline (crisis support service) were provided at the beginning and end of the survey. Respondents gave electronic informed consent by clicking the "next" button at the bottom of the information page. The project had approval from the CQUniversity Human Research Ethics Committee; approval number H17/04-06.

The survey included standardized measures of bereavement severity, posttraumatic growth, and social support; questions about the use of memorials; and demographic questions about themselves and their pet. The estimated minimum sample size needed for a medium effect size (with an alpha level of $p = 0.05$ and statistical power of 0.80) was 94 (with three predictors) or 117 (with six predictors). The final sample size achieved exceeded the minimum required: 133. These respondents were predominantly female (95.5%): 127 women, five men, and one individual who self-identified as genderqueer. Respondents ranged from 18 to 75 years of age ($M = 37.27$, $SD = 13.66$) and had owned their pet from one month to 26.4 years ($M = 10.67$, $SD = 5.34$). Most of the participants lived with a partner (62.4%), owned a dog (61.7%), and had had a pet euthanized (60.9%). Losses occurred between three months and five years before survey completion, with the average length of time since the loss being 2.39 years ($SD = 1.53$). Table 1 summarizes other key parameters.

Table 1. Demographic characteristics of the participants ($n = 133$).

Characteristic	<i>n</i>	%	Characteristic	<i>n</i>	%
<i>Household Type</i>			<i>Type of Pet</i>		
Couple with children	48	36.1	Dog	82	61.7
Couple without children	35	26.3	Cat	28	21.1
Lone person	23	17.3	Horse	12	9.0
Single parent family	10	7.5	Other	6	4.5
Non-family group	10	7.5	Rodent	3	2.3
Other	7	5.3	Bird	2	1.5
<i>Gender</i>			<i>Cause of Loss</i>		
Female	127	95.5	Died from an illness	56	42.1
Male	5	3.8	Died from old age	42	31.6
Other	1	0.8	Other	19	14.3
<i>Euthanized</i>			Died in an accident	16	12.0
Yes	81	60.9			
No	52	39.1			

Measures

Pet Bereavement Questionnaire (PBQ): This was used to measure grief severity. The scale consists of 16 items, including three subscales that measure grief (seven items), anger (five items), and guilt (four items). The original scale uses a 4-point Likert-type scale for responses, ranging from 0 (Disagree Strongly) to 3 (Agree Strongly). However, for the current study, a 5-point scale was used, ranging from 0 (Disagree Strongly) to 4 (Agree Strongly) and including 2 (Neutral). According to James and Rocco (2007), a minimum of 5 points are needed on a Likert-type scale to analyze the data using parametric techniques. Additionally, a neutral option was necessary so that participants were not forced to select an option that may not represent their true belief. The original scale has good internal reliability, with Hunt and Padilla (2006), Eckerd, Barnett, and Jett-Dias (2016) and Lee and Surething (2013) reporting Cronbach's alphas ranging from 0.85 to 0.87. In the current study, reliability for the total score ($\alpha = 0.84$) and subscales (ranging from $\alpha = 0.68$ to 0.83) was good.

Grief Experience Questionnaire (GEQ): The Loss of Social Support subscale of this scale was used to measure disenfranchised grief. The GEQ was designed to measure changes in social support from friends, family, and significant others following a loss (Barrett & Scott, 1989). This scale assesses the degree to which the bereaved experiences aspects of disenfranchised grief such as avoidance, a lack of understanding and concern, and an unwillingness to listen (Barrett & Scott, 1989). The Loss of Social Support subscale includes five items, measured using a 5-point Likert-type scale, ranging from 1 (Never) to 5 (Almost Always). Two items were changed to improve consistency ("feel avoided by friends" was changed to "feel that your friends avoided you" and "neighbors and in-laws" to "friends and family"). This subscale has good internal consistency ($\alpha = 0.86$) in the literature (Barrett & Scott, 1989) and in the current study ($\alpha = 0.88$).

Posttraumatic Growth Inventory-Short Form (PTGI-SF): This was used to measure post-traumatic growth. The PTGI-SF was designed to assess perceived benefits resulting from coping with trauma (Tedeschi & Calhoun, 1996). It contains five subscales: Relating to Others, New Possibilities, Personal Strength, Spiritual Change, and Appreciation of Life. The scale includes 10 items (2 for each subscale), measured using a 6-point scale, ranging from 0 (I did not experience this change) to 5 (I experienced this change to a very great degree). The anchors were abbreviated to improve the readability of the online survey (e.g., 0 = Not at all). The short form was selected as it has good internal reliability for the total score (ranging from $\alpha = 0.86$ to 0.89) and subscales ($\alpha = 0.68$ to 0.84) (Cann et al., 2010b). In the current study, reliability was acceptable for the total score ($\alpha = 0.91$) and for four of the five subscales (ranging from $\alpha = 0.71$ to 0.82). Reliability for the fifth subscale, Spiritual Change, was low ($\alpha = 0.50$), likely because of the small number of items in the scale. Cronbach's alpha is affected by the length of a measure (Tavakol & Dennick, 2011).

Memorials: Participants were asked several questions about memorial use, including whether they had a funeral service, scattered or kept ashes, or posted about the loss on social media. Participants were also asked if they had framed or mounted any photos of their pet or kept any mementos such as objects (e.g., favorite toys) or other keepsakes (e.g., fur or imprints of paw prints).

Data Analysis

Data were analyzed using IBM SPSS 23.0 (Armonk, NY, USA). After data cleaning and removal of responses where loss had occurred outside the specified timeframe, the final sample included 133 respondents. Comparison with the other memorial variables indicated that a question about arranging memorials was unclear as many respondents had answered incorrectly. Therefore, the analysis excluded this item. The data met most of the relevant assumptions for parametric analysis.

Results

Pet Bereavement Questionnaire

The mean score on the grief subscale on the Pet Bereavement Questionnaire ($M = 3.06$, $SD = 0.66$) was much higher than the mean guilt ($M = 1.80$, $SD = 1.05$) and anger ($M = 1.37$, $SD = 0.78$) subscale scores. The grief subscale was negatively skewed, whereas the anger subscale was positively skewed. The guilt subscale was slightly bimodal, but quite evenly spread. The corollary is that most respondents experienced high levels of grief, the majority were not particularly angry, and while most experienced some degree of guilt, only a minority felt extremely guilty. These findings are consistent with results found by Hunt and Padilla (2006) when developing the PBQ.

Grief Experience Questionnaire

The mean score on the GEQ Loss of Social Support subscale ($M = 1.90$, $SD = 0.90$) was consistent with grief scores following a loss from cancer, AIDS, or suicide (Houck, 2007; ranging from $M = 1.57$ to 2.93).

Posttraumatic Growth Inventory-Short Form

Mean scores for the PTGI-SF were low ($M = 1.50$, $SD = 1.12$) compared with scores for growth following a variety of stressful events reported by Cann et al. (2010b) when developing the PTGI-SF (ranging from $M = 3.65$ to 5.96, $SD = 0.93$ to 1.23). Mean subscale scores (see Table 2) ranged from 0.94 ($SD = 1.12$) (Spiritual Change) to 1.94 ($SD = 1.47$) (Personal Strength).

Table 2. Means, standard deviations and ranges of the scales used ($n = 133$).

Scale	<i>M</i>	<i>SD</i>	Range	
			Scale	Actual
<i>Pet Bereavement Questionnaire</i>	2.21	0.61	0.00–4.00	0.94–3.75
Grief subscale	3.06	0.66	0.00–4.00	0.00–4.00
Guilt subscale	1.80	1.05	0.00–4.00	0.00–4.00
Anger subscale	1.37	0.78	0.00–4.00	0.00–4.00
<i>Grief Experience Questionnaire: Loss of Social Support Subscale</i>	1.90	0.90	1.00–5.00	1.00–5.00
<i>Number of Memorials</i>	3.15	1.38	0.00–6.00	0.00–6.00
<i>Posttraumatic Growth Inventory</i>	1.50	1.12	0.00–5.00	0.00–4.10
Relating to Others	1.59	1.38	0.00–5.00	0.00–5.00
New Possibilities	1.27	1.38	0.00–5.00	0.00–5.00
Personal Strength	1.94	1.47	0.00–5.00	0.00–5.00
Spiritual Change	0.94	1.12	0.00–5.00	0.00–4.50
Appreciation of Life	1.74	1.34	0.00–5.00	0.00–5.00

Table 3. Regression analysis summary for disenfranchised grief and grief severity predicting posttraumatic growth ($n = 133$).

Variable	<i>b</i>	95% CI		<i>SE</i>	<i>t</i>	<i>p</i>	<i>R</i> ²
		Lower	Upper				
<i>Total Posttraumatic Growth</i>							0.108
Constant	1.59	1.40	1.79	0.099	16.14	< 0.001	
Grief Severity	0.36	0.06	0.65	0.148	2.40	0.018	
Disenfranchised Grief	0.05	−0.19	0.30	0.123	0.44	0.664	
Interaction ^a	−0.54	−0.83	−0.25	0.148	−3.64	< 0.001	0.076
<i>Relating to Others</i>							0.083
Constant	1.69	1.45	1.93	0.122	13.82	< 0.001	
Grief Severity	0.27	−0.08	0.62	0.177	1.53	0.128	
Disenfranchised Grief	−0.09	−0.36	0.19	0.140	−0.61	0.542	
Interaction	−0.57	−0.89	−0.24	0.164	−3.45	0.001	0.056
<i>New Possibilities</i>							0.075
Constant	1.38	1.13	1.63	0.127	10.85	< 0.001	
Grief Severity	0.31	−0.11	0.74	0.214	1.46	0.147	
Disenfranchised Grief	0.01	−0.29	0.31	0.153	0.07	0.942	
Interaction	−0.57	−0.98	−0.16	0.207	−2.77	0.007	0.056
<i>Personal Strength</i>							0.067
Constant	2.06	1.80	2.32	0.130	15.88	< 0.001	
Grief Severity	0.15	−0.26	0.55	0.203	0.72	0.476	
Disenfranchised Grief	0.13	−0.20	0.45	0.164	0.76	0.446	
Interaction	−0.65	−0.99	−0.30	0.175	−3.70	< 0.001	0.063
<i>Spiritual Change</i>							0.110
Constant	1.03	0.83	1.23	0.102	10.06	< 0.001	
Grief Severity	0.45	0.11	0.79	0.172	2.62	0.010	
Disenfranchised Grief	0.08	−0.15	0.32	0.121	0.70	0.486	
Interaction	−0.47	−0.84	−0.10	0.188	−2.51	0.013	0.058
<i>Appreciation of Life</i>							0.108
Constant	1.82	1.58	2.05	0.119	15.26	< 0.001	
Grief Severity	0.60	0.20	1.00	0.202	2.98	0.003	
Disenfranchised Grief	0.13	−0.19	0.46	0.163	0.42	0.416	
Interaction	−0.44	−0.86	0.02	0.213	−2.06	0.042	0.035

Note: Grief severity and disenfranchised grief were mean centered before analysis. Coefficients are unstandardized; standardized coefficients are not recommended when analysis includes a moderating variable (Field, 2013).

^aGrief severity $\times$ Disenfranchised grief interaction effect.

Main Analysis

A multiple regression analysis was used to test the hypothesis of a relationship between disenfranchised grief and posttraumatic growth in bereaved pet owners. PROCESS v.2.15 (Hayes, 2013) was used to determine if grief severity was moderating the relationship between

Table 4. Regression analysis summary for the variables and interaction effects predicting posttraumatic growth ($n = 133$).

Variable	<i>b</i>	95% CI		<i>SE</i>	<i>t</i>	<i>p</i>
		Lower	Upper			
Constant	1.60	1.38	1.81	0.108	14.73	< 0.001
Disenfranchised Grief	0.08	−0.21	0.36	0.145	0.52	0.598
Number of Memorials	0.00	−0.19	0.19	0.096	−0.01	0.996
Memorials × DG	−0.09	−0.31	0.13	0.114	−0.77	0.445
Grief Severity	0.36	0.04	0.67	0.160	2.24	0.027
Memorials × Grief Severity	0.12	−0.19	0.42	0.153	0.75	0.454
DG × Grief Severity	−0.56	−0.97	0.14	0.209	−2.66	0.009
Memorials × DG × Grief Severity	0.01	−0.38	0.40	0.198	0.04	0.970

Note: $R^2 = 0.117$. DG = Disenfranchised Grief. Grief severity, disenfranchised grief, and the number of memorials were mean centered before analysis. Coefficients are unstandardized; standardized coefficients are not recommended when analysis includes a moderating variable (Field, 2013).

disenfranchised grief and growth. Testing revealed that the assumptions of independence, linearity, normality, and non-multicollinearity were met.

No significant main effect was found between disenfranchised grief and total posttraumatic growth ($b = 0.05$, 95% CI [−0.19, 0.30], $t = 0.44$, $p = 0.66$). Disenfranchisement was not a significant predictor of growth for any of the subscales (b 's −0.09 to 0.13, all p 's > 0.05; see Table 3 for test statistics). The model showed that there was a significant interaction between grief severity and disenfranchised grief ($b = −0.54$, 95% CI [−0.83, −0.25], $t = −3.64$, $p < 0.001$), indicating that grief severity was moderating the relationship between disenfranchised grief and posttraumatic growth. The effect size was small ($R^2 = 0.08$) but acceptable as a practically significant effect (Ferguson, 2009). There was also a significant interaction for all five posttraumatic growth subscales. Effect sizes were small but acceptable ($R^2 = 0.06$) for the factors Relating to Others, New Possibilities, Personal Strength, and Spiritual Change. However, the effect size for Appreciation of Life was very small, falling below the recommended minimum threshold of 0.04 (Ferguson, 2009).

Simple slopes analysis (using high [+1 *SD*], mean, and low [−1 *SD*]) values of the moderator and predictor) was used to examine the moderation effect. Results showed that when grief severity was high, there was a significant negative relationship between disenfranchised grief and posttraumatic growth ($b = −0.27$, 95% CI [−0.49, −0.05], $t = −2.47$, $p = 0.015$). However, when grief severity was low, there was a significant positive relationship between disenfranchised grief and posttraumatic growth ($b = 0.38$, 95% CI [0.01, 0.75], $t = 2.05$, $p = 0.042$). Similarly, there was a significant negative relationship between disenfranchisement and growth at high levels of grief severity for the subscales Relating to Others ($b = −0.43$, 95% CI [−0.68, −0.18], $t = −3.42$, $p = 0.001$) and New Possibilities ($b = −0.34$, 95% CI [−0.62, −0.05], $t = −2.36$, $p = 0.020$), and a positive relationship at low levels of grief severity for the subscale Personal Strength ($b = 0.52$, 95% CI [0.06, 0.98], $t = 2.23$, $p = 0.028$).

The hypothesis that there is a relationship between memorial use and posttraumatic growth was tested using multiple regression analysis, with the number of memorials, grief severity, disenfranchised grief, and the interaction effects as predictors. Results showed that the direct effect of number of memorials was not a significant predictor of posttraumatic growth ($b < 0.01$, 95% CI [−0.19, 0.19], $t = −0.01$, $p = 0.996$) (see Table 4). As expected, the interaction

Table 5. Regression analysis summary for type of memorial predicting posttraumatic growth ($n = 133$).

Variable	<i>b</i>	95% CI		<i>SE</i>	β	<i>p</i>
		Lower	Upper			
Constant	1.60	0.20	3.01	0.711		0.026
Scattered or Kept Ashes	0.08	−0.33	0.48	0.205	0.03	0.702
Funeral Service	0.12	−0.37	0.61	0.246	0.04	0.636
Posted on Social Media	0.25	−0.20	0.69	0.223	0.10	0.274
Kept an Object as a Memento	−0.12	−0.61	0.38	0.249	−0.05	0.640
Kept Fur etc. as a Memento	−0.23	−0.66	0.19	0.215	−0.10	0.281
Framed or Mounted a Photo	−0.18	−0.62	0.26	0.224	−0.08	0.420

Note: $R^2 = 0.031$.

between disenfranchised grief and grief severity was significant. However, no other interactions were significant, indicating that neither grief severity, disenfranchised grief, nor the interaction between grief severity and disenfranchisement were moderating the relationship between the number of memorials and posttraumatic growth. A separate regression analysis showed no significant relationships between the type of memorial and posttraumatic growth (see Table 5).

Discussion

The purpose of this study was to better understand the grief experience following the loss of a pet by examining the relationship between disenfranchised grief and posttraumatic growth. Most research in this field focuses on the bereavement experience following the loss of a human family member and it was expected that growth after pet loss would differ because pet loss is a form of disenfranchised grief. Overall, scores on the Loss of Social Support subscale of the Grief Experience Questionnaire were moderate, suggesting that many respondents felt somewhat socially isolated following their loss. The mean score was slightly higher than scores found by Harwood, Hawton, Hope, and Jacoby (2002), who used the GEQ to measure bereavement following the loss of a human loved one by suicide. However, scores on the Posttraumatic Growth Inventory-Short Form were quite low, suggesting that most respondents only experienced a small amount of change. By comparison, Cann et al. (2010b) used the PTGI-SF to assess posttraumatic growth following a variety of traumatic events (e.g., death of a loved one, divorce, medical issues, and assault), finding that respondents experienced a moderate amount of posttraumatic growth.

Consistent with the hypothesis that disenfranchised grief predicts posttraumatic growth, results indicated that disenfranchised grief was negatively related to posttraumatic growth when grief severity was high. This finding is consistent with the literature. Using a different approach, Wong et al. (2015) also found that most respondents who reported posttraumatic growth had social support from friends, family, or counselors, whereas those who had difficulty processing their grief tended to avoid others and did not receive social support or experience growth. Additionally, Morris et al. (2005) assessed the relationship between trauma severity and growth, finding that both objective and subjective trauma ratings predicted posttraumatic growth. Morris et al. (2005) reported that severe trauma ratings were related to higher levels of growth, which is consistent with the findings of the current study as respondents with high levels of grief severity and low levels of disenfranchised grief experienced the most growth.

Our study also revealed a positive relationship between disenfranchised grief and post-traumatic growth when grief severity was low. In this group, individuals with low levels of disenfranchised grief experienced less growth than those who were more disenfranchised. This finding is partially consistent with Tedeschi and Calhoun's (2004) functional-descriptive model, which suggests that posttraumatic growth will not occur unless assumptive beliefs are challenged. Instead, pre-existing beliefs provide context for the loss, mitigating emotional distress. Respondents who experienced a low level of grief severity and little disenfranchisement probably retained their pre-existing beliefs and accepted the loss without growth. Respondents with a low level of grief severity but high levels of disenfranchisement achieved a moderate amount of growth. This result differs from Tedeschi and Calhoun's (2004) finding that self-disclosure and social support were helpful when processing a trauma into growth. Further, adverse sociocultural influences (including disenfranchised grief), can be detrimental to the grieving process and may affect posttraumatic growth. However, additional factors such as personality differences or resilience may affect the long-term impact of the loss (Calhoun et al., 2010). It is possible that other characteristics (unmeasured in the current study) mediate the relationship between low grief severity and high disenfranchisement.

There was a significant interaction between disenfranchised grief and grief severity for the subscales Relating to Others, New Possibilities, and Personal Strength. Results indicate that disenfranchised grief was negatively related to the subscales Relating to Others and New Possibilities at high levels of grief severity. This relationship is consistent with the qualitative findings of Packman et al. (2017), who reported that while bereaved pet owners commonly endorsed items on the Relating to Others subscale, very few indicated that they felt socially supported. Packman et al. (2017) found that items on the New Possibilities subscale were the least endorsed. Comments tended to focus on caring for their other pets, and few respondents mentioned new possibilities for themselves or other humans, possibly due to disenfranchisement and isolation from other people. Comparatively, the current study found a positive relationship between disenfranchised grief and Personal Strength when grief severity was low. Overall, these findings suggest that high levels of grief severity and disenfranchisement inhibit growth, particularly in areas that involve interpersonal relationships and opportunities, whereas low levels of grief severity and disenfranchised grief tend to promote growth in areas involving self-reliance and coping with difficulties. The least growth was reported on the Spiritual Change subscale, which may relate to differences between Australia and other cultures, as Morris et al. (2005) also found very low levels of spiritual growth in an Australian sample, and Bussolari et al. (2017) found that Spiritual Change was endorsed much more commonly in the United States than in other cultures. Additionally, reliability for this subscale was low ($\alpha = 0.50$) and may have influenced the results.

There was no significant relationship between the number of memorials or type of memorials and posttraumatic growth. Conversely, Castle and Phillips (2003) reported that most participants experienced positive gains (such as changed priorities, acceptance, and re-evaluation of relationships) from memorial activities. However, they examined a more extensive variety of memorial activities than those included in the current study, including keeping a journal, singing, painting, and speaking to the deceased. Ullrich and Lutgendorf (2002) examined the effect of journaling following trauma and found that respondents who wrote about cognitions and emotions were more aware of the positive benefits of the trauma than a control group and those who only wrote about emotions. Therefore, it seems likely that certain memorial activities are more strongly related to posttraumatic growth than others.

Limitations

The current study had several limitations, including a predominantly female sample. While response rates for survey-based research tend to be lower for males than females (Smith, 2008), the extremely skewed gender sample in the current study limits the generalizability of results. While Wrobel and Dye (2003) found that females reported higher frequencies of grief symptoms than males, self-reported symptom frequency is unlikely to capture the underlying severity of grief or the influence of societal norms. Doka and Martin (2010) argue that gender-stereotyped behavior can create a dissonance between rigid definitions of masculinity and the need to grieve, resulting in self-disenfranchisement and the suppression of grief reactions. Further research is needed to explore the influence of societal gender norms on disenfranchised grief and posttraumatic growth.

Another limitation of the current study is the online methodology with self-selected respondents. Online research can exclude certain groups, including respondents who are elderly or of low socioeconomic status. However, Tolstikova and Chartier (2010) compared a self-selected online group and a self-selected paper-and-pencil group and had results consistent with those in the current study: both the online and paper-and-pencil groups were mostly female and there were few significant differences between the two groups.

Future Research Directions and Conclusion

As hypothesized, the current study shows that disenfranchised grief can impact posttraumatic growth following the loss of a pet. These findings highlight the complicated relationship between the domains of posttraumatic growth, grief intensity, and disenfranchised grief. Future research should focus on expanding our understanding of the role of grief severity in this relationship, as well as explore additional factors such as personality, resilience, attachment, and cognitive processing. The present study shows that there is a relationship between disenfranchised grief and the use of memorials but there is not a relationship between memorial use and posttraumatic growth. Future research should include an in-depth exploration of the benefits of a wider variety of rituals and memorials following the loss of a companion animal and their relationship with disenfranchised grief and posttraumatic growth. The results of this study provide insight into the bereavement experience following pet loss, enhancing our knowledge of the adverse consequences of disenfranchised grief on the psychological wellbeing and growth of bereaved pet owners.

Conflicts of Interest

The authors state that there are no conflicts of interest.

References

- Adams, C. L., Bonnett, B. N., & Meek, A. H. (1999). Owner response to companion animal death: Development of a theory and practical implications. *The Canadian Veterinary Journal*, 40, 33–39. Retrieved from <https://www.canadianveterinarians.net/science-knowledge/cvj>.
- Adrian, J. A. L., & Stitt, A. (2017). Pet loss, complicated grief, and post-traumatic stress disorder in Hawaii. *Anthrozoös*, 30, 123–133. doi:10.1080/08927936.2017.1270598.
- Allen, K., Blascovich, J., & Mendes, W. B. (2002). Cardiovascular reactivity and the presence of pets, friends, and spouses: The truth about cats and dogs. *Psychosomatic Medicine*, 64, 727–739. doi:10.1097/01.PSY.0000024236.11538.41.
- Animal Medicines Australia. (2016). *Pet ownership in Australia: 2016*. Retrieved from <http://animalmedicinesaustralia.org.au/pet-report/>.

- Aydin, N., Krueger, J. L., Fischer, J., Hahn, D., Kastenmüller, A., Frey, D., & Fischer, P. (2011). "Man's best friend:" How the presence of a dog reduces mental distress after social exclusion. *Journal of Experimental Social Psychology*, 48, 446–449. doi:10.1016/j.jesp.2011.09.011.
- Barrett, T. W., & Scott, T. B. (1989). Development of the Grief Experience Questionnaire. *Suicide and Life-Threatening Behavior*, 19, 201–215. doi:10.1111/j.1943-278X.1989.tb01033.x.
- Bonanno, G. A., Papa, A., Lalande, K., Zhang, N., & Noll, J. G. (2005). Grief processing and deliberate grief avoidance: A prospective comparison of bereaved spouses and parents in the United States and the People's Republic of China. *Journal of Consulting and Clinical Psychology*, 73, 86–98. doi:10.1037/0022-006X.73.1.86.
- Bowlby, J. (1982). *Attachment: Attachment and loss. Volume 1* (2nd ed.). New York, NY: Basic Books.
- Bussolari, C., Habarth, J., Kimpara, S., Katz, R., Carlos, F., Chow, A., . . . Packman, W. (2017). Posttraumatic growth following the loss of a pet: A cross-cultural comparison. *OMEGA: Journal of Death and Dying*, 75, 337–359. doi:10.1177/0030222817690403.
- Calhoun, L. G., Tedeschi, R. G., Cann, A., & Hanks, E. A. (2010). Positive outcomes following bereavement: Paths to posttraumatic growth. *Psychologica Belgica*, 50, 125–143. doi:10.5334/pb-50-1-2-125.
- Cann, A., Calhoun, L. G., Tedeschi, R. G., Kilmer, R. P., Gil-Rivas, V., Vishnevsky, T., & Danhauer, S. C. (2010a). The Core Beliefs Inventory: A brief measure of disruption in the assumptive world. *Anxiety, Stress & Coping*, 23, 19–34. doi:10.1080/10615800802573013.
- Cann, A., Calhoun, L. G., Tedeschi, R. G., Taku, K., Vishnevsky, T., Triplett, K. N., & Danhauer, S. C. (2010b). A short form of the Posttraumatic Growth Inventory. *Anxiety, Stress & Coping*, 23, 127–137. doi:10.1080/10615800903094273.
- Castle, J., & Phillips, W. L. (2003). Grief rituals: Aspects that facilitate adjustment to bereavement. *Journal of Loss and Trauma*, 8, 41–71. doi:10.1080/15325020305876.
- Chur-Hansen, A. (2010). Grief and bereavement issues and the loss of a companion animal: People living with a companion animal, owners of livestock, and animal support workers. *Clinical Psychologist*, 14, 14–21. doi:10.1080/13284201003662800.
- Chur-Hansen, A. (2011). Cremation services upon the death of a companion animal: Views of service providers and service users. *Society & Animals*, 19, 248–260. doi:10.1163/156853011X578910.
- Cobb, A. R., Tedeschi, R. G., Calhoun, L. G., & Cann, A. (2006). Correlates of posttraumatic growth in survivors of intimate partner violence. *Journal of Traumatic Stress*, 19, 895–903. doi:10.1002/jts.20171.
- Cordaro, M. (2012). Pet loss and disenfranchised grief: Implications for mental health counseling practice. *Journal of Mental Health Counseling*, 34, 283–294. doi:10.17744/mehc.34.4.41q0248450t98072.
- Corr, C. A. (1999). Enhancing the concept of disenfranchised grief. *OMEGA: Journal of Death and Dying*, 38, 1–20. doi:10.2190/LD26-42A6-1EAV-3MDN.
- Doka, K. J. (1999). Disenfranchised grief. *Bereavement Care*, 18, 37–39. doi:10.1080/02682629908657467.
- Doka, K. J. (2002). Disenfranchised grief. In K. J. Doka (Ed.), *Living with grief: Loss in later life* (pp. 159–168). Washington, DC: The Hospice Foundation of America.
- Doka, K. J., & Martin, T. L. (2010). *Grieving beyond gender: Understanding the ways men and women mourn*. New York, NY: Routledge.
- Eckerd, L. M., Barnett, J. E., & Jett-Dias, L. (2016). Grief following pet and human loss: Closeness is key. *Death Studies*, 40, 275–282. doi:10.1080/07481187.2016.1139014.
- Ferguson, C. J. (2009). An effect size primer: A guide for clinicians and researchers. *Professional Psychology: Research and Practice*, 40, 532–538. doi:10.1037/a0015808.
- Harwood, D., Hawton, K., Hope, T., & Jacoby, R. (2002). The grief experiences and needs of bereaved relatives and friends of older people dying through suicide: A descriptive and case-control study. *Journal of Affective Disorders*, 72, 185–194. doi:10.1016/S0165-0327(01)00462-1.
- Hayes, A. F. (2013). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. New York, NY: Guilford Press.
- Houck, J. A. (2007). A comparison of grief reactions in cancer, HIV/AIDS, and suicide bereavement. *Journal of HIV/AIDS & Social Services*, 6(3), 97–112. doi:10.1300/J187v06n03_07.
- Hunt, M., & Padilla, Y. (2006). Development of the Pet Bereavement Questionnaire. *Anthrozoös*, 19, 308–324. doi:10.2752/089279306785415493.
- James, C., & Rocco, J. P. (2007). Ten common misunderstandings, misconceptions, persistent myths and urban legends about Likert scales and Likert response formats and their antidotes. *Journal of Social Sciences*, 3, 106–116. doi:10.3844/jssp.2007.106.116.

- Kemp, H. R., Jacobs, N., & Stewart, S. (2016). The lived experience of companion-animal loss: A systematic review of qualitative studies. *Anthrozoös*, 29, 533–557. doi:10.1080/08927936.2016.1228772.
- Kenney, E. (2004). Pet funerals and animal graves in Japan. *Mortality*, 9, 42–60. doi:10.1080/13576270410001652532.
- Lee, S. A., & Surething, N. A. (2013). Neuroticism and religious coping uniquely predict distress severity among bereaved pet owners. *Anthrozoös*, 26, 61–76. doi:10.2752/175303713X13534238631470.
- Morris, B. A., Shakespeare-Finch, J., Rieck, M., & Newbery, J. (2005). Multidimensional nature of posttraumatic growth in an Australian population. *Journal of Traumatic Stress*, 18, 575–585. doi:10.1002/jts.20067.
- Neimeyer, R. A., & Jordan, J. R. (2002). Disenfranchisement as empathic failure: Grief therapy and the co-construction of meaning. In K. J. Doka (Ed.), *Disenfranchised grief: New directions, challenges, and strategies for practice* (pp. 95–117). Champaign, IL: Research Press.
- Packman, W., Bussolari, C., Katz, R., Carmack, B. J., & Field, N. P. (2017). Posttraumatic growth following the loss of a pet. *OMEGA: Journal of Death and Dying*, 75, 337–359. doi:10.1177/0030222816663411.
- Packman, W., Carmack, B. J., Katz, R., Carlos, F., Field, N. P., & Landers, C. (2014). Online survey as empathic bridging for the disenfranchised grief of pet loss. *OMEGA: Journal of Death and Dying*, 69, 333–356. doi:10.2190/OM.69.4.a.
- Parkes, C. M. (1971). Psycho-social transitions: A field for study. *Social Science & Medicine*, 5, 101–115. doi:10.1016/0037-7856(71)90091-6.
- Sable, P. (1995). Pets, attachment, and well-being across the life cycle. *Social Work*, 40, 334–341. doi:10.1093/sw/40.3.334.
- Sharkin, B. S., & Knox, D. (2003). Pet loss: Issues and implications for the psychologist. *Professional Psychology: Research and Practice*, 34, 414–421. doi:10.1037/0735-7028.34.4.414.
- Smith, W. G. (2008). *Does gender influence online survey participation?: A record-linkage analysis of university faculty online survey response behavior*. Retrieved from <https://eric.ed.gov/?id=ED501717>.
- Tavakol, M., & Dennick, R. (2011). Making sense of Cronbach's alpha. *International Journal of Medical Education*, 2, 53–55. doi:10.5116/ijme.4dfb.8dfd.
- Tedeschi, R. G., & Calhoun, L. G. (1996). The Posttraumatic Growth Inventory: Measuring the positive legacy of trauma. *Journal of Traumatic Stress*, 9, 455–471. doi:10.1002/jts.2490090305.
- Tedeschi, R. G., & Calhoun, L. G. (2004). Posttraumatic growth: Conceptual foundations and empirical evidence. *Psychological Inquiry*, 15, 1–18. doi:10.1207/s15327965pli1501_01.
- Tolstikova, K., & Chartier, B. (2010). Internet method in bereavement research: Comparison of online and offline surveys. *OMEGA: Journal of Death and Dying*, 60, 327–349. doi:10.2190/OM.60.4.b.
- Tzivan, L., Friger, M., & Kushnir, T. (2015). Associations between stress and quality of life: Differences between owners keeping a living dog or losing a dog by euthanasia. *PLoS ONE*, 10(3), 1–15. doi:10.1371/journal.pone.0121081.
- Ulrich, P. M., & Lutgendorf, S. K. (2002). Journaling about stressful events: Effects of cognitive processing and emotional expression. *Annals of Behavioral Medicine*, 24, 244–250. doi:10.1207/S15324796ABM2403_10.
- Wolchik, S. A., Cox, S., Tein, J. Y., Sandler, I. N., & Ayers, T. S. (2009). Six-year longitudinal predictors of posttraumatic growth in parentally bereaved adolescents and young adults. *OMEGA: Journal of Death and Dying*, 58, 107–128. doi:10.2190/OM.58.2.b.
- Wong, P. W. C., Lau, K. C. T., Liu, L. L., Yuen, G. S. N., & Wing-Lok, P. (2015). Beyond recovery: Understanding the postbereavement growth from companion animal loss. *OMEGA: Journal of Death and Dying*, 75, 103–123. doi:10.1177/0030222815612603.
- Wrobel, T. A., & Dye, A. L. (2003). Grieving pet death: Normative, gender, and attachment issues. *OMEGA: Journal of Death and Dying*, 47, 385–393. doi:10.2190/QYV5-LLJ1-T043-U0F9.